

# 8 - Reserve Prices

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## Reserve Prices

A price  $r > 0$ , such the object is not sold if the auction price is less than  $r$ .

## Reserve Prices in the SPA

→ No bidder with value  $v < r$  will participate  $\Rightarrow$  Such a bidder cannot make a positive profit.

$\Rightarrow$  Only bidders with values  $v \geq r$  will participate.

$\Rightarrow$  It is still a weakly dominant strategy to bid your own value.

For a bidder with value  $v = r$ , the expected payment is:

$$G_1^{(n-1)}(r) r.$$

What about for bidders  $v > r$ , the expected payment is

$$m^{\#}(v) = G_1^{(n-1)}(r) r + \int_r^v y g_1^{(n-1)}(y) dy$$

## Reserve Prices in the FPA

- No bidder with value  $v < r$  can make a positive profit.
- If  $\beta^\pi(v)$  is a symmetric eq. of the FPA, we must have that  $\beta^\pi(r) = \underline{r}$ .
- If  $v > r$ ,

$$\beta^\pi(v) = E[\max\{Y_i^{(n-1)}, r\} \mid Y_i^{(n-1)} \leq v]$$

$$= \frac{1}{G_i^{(n-1)}(v)} \cdot \int_0^v \max\{y, r\} g_i^{(n-1)}(y) dy$$

$$= \frac{1}{G_i^{(n-1)}(v)} \int_0^r g_i^{(n-1)}(y) dy + \frac{1}{G_i^{(n-1)}(v)} \int_r^v y g_i^{(n-1)}(y) dy$$

$$= \frac{r G_i^{(n-1)}(r)}{G_i^{(n-1)}(v)} + \frac{1}{G_i^{(n-1)}(v)} \int_r^v y g_i^{(n-1)}(y) dy$$

$$m^\pi(v, r) = r G_i^{(n-1)}(r) + \int_r^v y g_i^{(n-1)}(y) dy \quad v \geq r$$

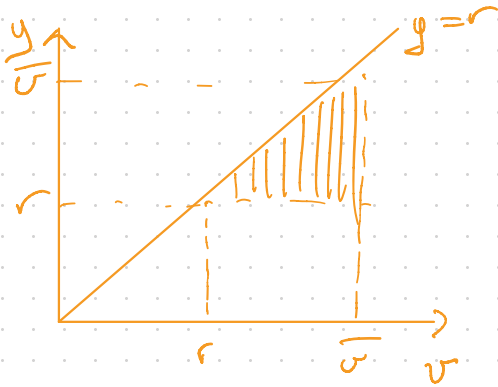
$$m^A(v, r) = \left( r G_1^{(n-1)}(r) + \int_r^v y g_1^{(n-1)}(y) dy \right) \frac{1}{\int_0^v r g_1^{(n-1)}(r) dr} A \in [I, II]$$

$$E[m^A(V, r)] = \int_r^{\bar{v}} \left\{ \underline{r G_1^{(n-1)}(r)} + \underline{\int_r^v y g_1^{(n-1)}(y) dy} \right\} f(v) dv$$

$$= r G_1^{(n-1)}(r) \int_r^{\bar{v}} f(v) dv + \int_r^{\bar{v}} \int_r^v y g_1^{(n-1)}(y) dy f(v) dv$$

$$= r G_1^{(n-1)}(r) (F(\bar{v}) - F(r)) + \int_r^{\bar{v}} y g_1^{(n-1)}(y) (1 - F(y)) dy$$

$$= r G_1^{(n-1)}(r) (1 - F(r)) + \int_r^{\bar{v}} y g_1^{(n-1)}(y) (1 - F(y)) dy$$



## The auctioneer's revenue:

- Suppose the seller attaches a value  $v_0 \in [0, \bar{v})$  to the object
- The seller won't use  $r < v_0$
- The expected revenue of  $r \geq v_0$  is

$$\begin{aligned}\Pi_0 &= n \mathbb{E}[m^A(V, r)] + \underbrace{F(r)^n v_0}_{G_1^{(n)}(r)} \\ &= n r G_1^{(n-1)}(r) - n r G_1^{(n-1)}(r) F(r) + n \int_r^{\bar{v}} y (1-F(y)) g_1^{(n-1)}(y) dy + F(r)^n v_0\end{aligned}$$

To find the optimal  $r^*$ , we need to find  $\frac{d\Pi_0}{dr}$

$$\Pi_0 = n r G_1^{(n-1)}(r) - n r G_1^{(n-1)}(r) F(r) + n \int_r^{\bar{v}} y (1-F(y)) g_1^{(n-1)}(y) dy + F(r) \bar{v}_0$$

$$\Rightarrow \frac{d\Pi_0}{dr} = n G_1^{(n-1)}(r) - n G_1^{(n-1)}(r) F(r) - n r G_1^{(n-1)}(r) f(r) + n G_1^{(n-1)}(r) f(r) \bar{v}_0$$

$$= n [1 - F(r) - r f(r) + f(r) \bar{v}_0] G_1^{(n-1)}(r) \cdot \frac{1 - F(r)}{1 - F(r)}$$

Define the Hazard Rate of  $F$  :  $\lambda(x) = \frac{f(x)}{1 - F(x)}$

$$= n [1 - r \lambda(r) + \bar{v}_0 \lambda(r)] (1 - F(r)) G_1^{(n-1)}(r)$$

$$= n [1 - (r - \bar{v}_0) \lambda(r)] (1 - F(r)) G_1^{(n-1)}(r)$$

FOC:

$$n [1 - (r^* - \sigma_0) \lambda(r^*)] \underbrace{(1 - F(r^*)) G_1^{(n-1)}(r^*)}_{> 0} = 0$$

$$\Leftrightarrow 1 - (r^* - \sigma_0) \lambda(r^*) = 0$$

$$\boxed{r^* - \frac{1}{\lambda(r^*)} = \sigma_0}$$

$$\text{If } \sigma_0 = 0 \Rightarrow r^* = \frac{1}{\lambda(r^*)}$$

Example:  $U[0,1]$

$$\lambda(x) = \frac{1}{1-x} \Rightarrow r^* = 1 - r^* \Rightarrow r^* = 1/2$$